Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years shipping **production web systems**, **Postgres/SQL** backends, tested releases, and **AI-assisted workflows** across fintech, operations, and customer-facing platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Built Azure-backed AI education platform with normalized SQL schemas, row-level isolation, and indexed dashboards, reducing latency from 1.2 seconds to 350 milliseconds for 500+ campus users across releases.
- Reworked FastAPI background jobs with explicit states, failure isolation, and SQL-backed status tracking, processing 1K+ AI jobs while tripling throughput under concurrent instructor workloads, tickets, and reviews.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Implemented React and Flask grading workflows for uploads, rubrics, history, and model selection, supporting 150+ instructors and 10K+ daily requests through production REST APIs, dashboards, and fixes.
- Designed concurrent file-ingestion services with ThreadPoolExecutor, OCR, and multimodal AI review, reducing feedback turnaround 60% while keeping batch processing testable, observable, and responsive for instructors and QA.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Troubleshot large-scale API integrations using OOP patterns, SQL workflows, AWS Glue, and S3, improving response times 30% while reducing technical debt 20% across release cycles and defects.
- Engineered cross-platform microservices with Jenkins CI/CD, VM automation, and 20+ PyTest, Mockito, unit, and integration suites while tracking defects through Agile sprint delivery with QA teammates consistently.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed React/Node resource-management workflows with REST APIs, WebSockets, and modular services, synchronizing 2M+ records across locations with 33% fewer conflicts during rollout cycles and review tickets consistently.
- Implemented CRM modules for authentication, resources, billing, leads, permissions, and notifications, using Jenkins CI/CD and Docker staging to cut deployment time 50% safely across releases and fixes.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built backend APIs for booking, slot selection, reservation tracking, coupons, and payment-status updates across 50K+ records, delivering sub-200-millisecond responses for client-facing workflows and support reviews reliably remotely.
- Mapped 12+ integration points across Flask APIs, Firebase synchronization, webhooks, payments, and cross-platform delivery, reducing design errors 25% through early stakeholder architecture reviews and documentation handoffs collaboratively.

Skills

Full-Stack: JavaScript, TypeScript, Python, Java, Ruby, Ruby on Rails, Hotwire, Turbo, Stimulus, React, Next.js, Node.js
Backend & Data: Flask, FastAPI, REST APIs, Webhooks, PostgreSQL, SQL, MongoDB, Firebase, Redis, AWS, S3, Docker
AI & Product: AI Agents, GitHub Copilot, Claude Code, LLM Workflows, OCR, Prompt Review, Human-in-the-Loop Review, Edge Cases
Quality & Startup Delivery: Unit Testing, Integration Testing, PyTest, Mockito, CI/CD, Jenkins, GitHub Actions, Git, Agile/Scrum, Code Reviews, Defect Tracking

Projects

BrokerOps Agent Console | *Ruby on Rails, Hotwire, PostgreSQL, AWS, AI Agents*

- Designed an insurance-brokerage workflow concept with Rails-style models, Hotwire screens, Postgres quote records, AWS deployment planning, AI-agent intake assistance, human review checkpoints, and test cases for edge conditions.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid, WebSockets, Kubernetes*

- Built a fintech platform supporting secure bank linking, real-time transaction updates, tokenized access, and filtering across 1M+ transactions, using Next.js, TypeScript, MongoDB, Plaid, and WebSockets reliably online.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker*

- Built an AI-powered SQL workspace supporting 10+ databases, schema exploration, query generation, and validation, improving data-access productivity through Java services, TypeScript modules, and Docker-based local deployment workflows.